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MILITARY SPECIFICATION

SYRINGE, IRRIGATING, SURGICAL, CATHETER TIP, DISPOSABLE, 50 cc (2 Fl. Oz.)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a sterile, disposable, plastic irrigating syringe with a catheter tip for use with irrigating catheters and intubating tubes and in gastric sampling, inflation of Foley catheters during cystoscopy, and irrigating wet dressings.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636 - Boxes, Shipping, Fiberboard.

MILITARY

MIL-S-36586 - Sterilization Test Strip Set, Bacterial Spore.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6515

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(Copies of specification, standards, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

THE UNITED STATES PHARMACOPEIAL CONVENTION, INC.

The Pharmacopeia of the United States of America (USP).

(Application for copies shall be addressed to the Mack Publishing Company, Easton, Pennsylvania 18042).

U.S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE;
FOOD AND DRUG ADMINISTRATION

Federal Food, Drug, and Cosmetic Act and General Regulations for its Enforcement.

(Application for copies should be addressed to the Food and Drug Administration, U.S. Department of Health, Education and Welfare, Washington, DC 20204.)

3. REQUIREMENTS

3.1 Material. All material shall comply with the applicable requirements of the Federal Food, Drug and Cosmetic Act. All plastic components of the syringe shall comply with the Pharmacopeia of the United States (USP) requirements for Class III plastic containers.

3.1.2 Barrel. The syringe barrel shall be fabricated from rigid, transparent, virgin or selected regrind polypropylene plastic. The regrind shall be free from graduation filler and any other adulterants. The barrel shall be considered to be transparent if the scale can be seen in daylight by the unaided eye when viewed through the empty barrel from the side opposite the scale.

3.1.3 Plunger. The plunger rod shall be made of suitable polypropylene plastic. The plunger tip shall be fabricated from fresh stock non-blooming, non-leaching, non-toxic, age-resistant, inert, natural rubber or neoprene.

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3.1.4 Lubricant. Shall be inert, non-toxic, silicone compound suitable for the specific use. It shall also serve as a siliconizing coating on all interior surfaces of the syringe that come in contact with the plunger tip. The silicone compound shall be a pharmaceutical grade.

3.2 Style, design, and construction. Shall be a sterile, pyrogen-free, non-toxic, disposable, irrigating syringe. The syringe shall consist of a plunger, barrel and tip protector, or, as an alternate, an immediate container so designed as to facilitate dispensing the syringe by sterile technique.

3.2.1 Barrel. The syringe barrel shall be fabricated from plastic as specified in 3.1.2. The barrel shall consist of a tube closed at one end except for an orifice and irrigating tip, and open at the other end to receive the plunger. The tip shall be concentric to the axis of the barrel. The barrel dimensions shall be as listed in table I.

TABLE I. Barrel dimensions (inches) and graduations.

Size	Overall length	Outside 1/ diameter	Wall thickness	Inside diameter at tip	Subgrad-uations	Main graduations	Numbering
7 (50 cc)	6.62 to 7.34	1.154 to 1.215	.041 min	.144 to .214	5 cc (1/4 oz)	10 cc (1/2 oz)	10, 20, 30, 40, 50 cc (1/2, 1, 1-1/2, 2 oz)

1/ Near distal end.

The length of the tip shall be 1.24 to 1.81 inches long and shall taper from 0.440 ± .010 at the proximal end to .250 ± .010 inch at the distal end. The wall at the open end of the barrel shall extend radially outward to form a flange of suitable size and shape to provide finger grip and antiroll features. The inside of the barrel shall be finished to a very fine, smooth and uniform surface substantially free from pits, grind marks, bubbles, mold marks, flash, or other defects that would affect the tightness and smoothness of fit with the plunger. The inside surface of the barrel shall be lightly coated with the silicone lubricant specified in 3.1.4. The inside diameter of the barrel shall be such as to provide an accuarate fit with the plunger tip, resulting in a fluid tight fit (see 3.4 and 3.5), and yet permitting the plunger to move freely throughout its excursion in the barrel. The ungraduated portion of the barrel at the proximal end shall be of sufficient length to permit the plunger tip to be aligned with the top graduation mark, while the barrel is encompassing the entire rubber tip, but shall be no greater then one-third of the overall length of the syringe barrel.

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3.2.1.1 Graduations and other markings. The outside surfaces of the barrel shall be marked with a graduation scale and the name or registered trademark of the contractor. It is desirable that the graduation scale shall be applied to the barrel in such a manner as to be approximately in line with the flattened side of the flange. The zero (0) mark of the scale shall register precisely with the distal ring of the plunger tip, when plunger is pushed to the bottom of the barrel. The scale and numbering shall be in cubic centimeters and ounces as specified in table I. Main graduation marks shall be at least 33-1/3 percent longer than subgraduation marks. No number is required at the zero (0) mark. The markings shall be legible and of a dark, contrasting color, easily readable.

3.2.1.1.1 Alcohol resistance of graduation markings. The graduation markings shall withstand vigorous rubbing of an alcohol moistened cotton swab and remain legible when tested according to 4.4.7.

3.2.2 Plunger. The plunger shall be either hollow or solid, fabricated from the material specified in 3.1.3. The distal end of the plunger shall be shaped to permit attachment of a plunger tip. The proximal end of the plunger shall extend well out of the barrel, and shall be shaped to form a knob of a thickness and diameter appropriate to the size of the syringe, and shaped to allow a suitable finger grip for usage. The distal end of each plunger with its attached plunger tip shall accurately fit its individual barrel so as to make the syringe successfully withstand the fluid tightness requirements specified in 3.4, permit the plunger and plunger tip to draw liquids, and yet move freely and smoothly throughout its excursion in the barrel. The plunger tip shall be securely attached to the plunger rod so as not to become loose or detached when in use. The tip shall be inside of the barrel when the plunger is pushed all the way in. In addition, the tip shall be designed, e.g., beveled, in order to permit easy and efficient removal of air bubbles from fluid path. The plunger tip shall be properly treated with the silicone lubricant specified in 3.1.4. An excess of lubricant that detracts from appearance, interferes with visibility of plunger tip ring, or can be caused to form pools or droplets, or can be washed out with the solution being used, shall be cause for rejection. Graduation markings, denoting correct, accurate capacity, shall register precisely with the first ring touching the sides of the barrel at the plunger tip distal end. Tip shall be of a permanent, suitable color so that the first ring at the plunger tip distal end, touching sides of the barrel, shall be clearly visible and shall accurately indicate through the barrel wall. Each plunger shall have a suitable solid thumbrest at its proximal end. Thumbrest shall be perpendicular to the centerline of the plunger and may be flat or slightly curved. As an alternate, thumbrest may have a centered hole which shall not exceed 3/8 inch in diameter. Thumbrest shall be fabricated from material as specified in 3.1.3 for the plunger rod.

3.2.2.1 Inertness to medication. The plunger tip shall show no evidence of a reaction with a high pH buffer solution, when tested according to 4.4.8.

3.2.3 Tip protector. Each syringe shall be provided with a protector fabricated from a suitable rigid, non-toxic, high grade plastic. The protector shall fit securely over the syringe tip and shall maintain the steriltiy of the syringe tip. The protector shall be so secured on the syringe as to remain in place during shipment and normal handling. As an alternate, each syringe in its entirety may be packaged in a rigid, plastic, tamperproof container designed to facilitate dispensing the syringe by sterile technique and to maintain the sterility of the syringe until the container is damaged or open.

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3.3 Accuracy. When teated as specified in 4.4.1, syringes shall deliver plus or minus 5 percent of the capacity indicated at each graduation mark tested.

3.4 Pressure test. The syringe shall be tested for fluid tightness between the plunger and barrel, as specified in 4.4.2.1. During the first test, there shall not be enough water to pass the plunger to produce a falling drop. After each successive test, any water that may have passed the plunger shall be wiped away and the next test applied during which there shall not be enough water to pass the plunger to produce a falling drop.

3.5 Suction test. There shall be no air leakage as revealed by the suction test as specified in 4.4.2.2 or 4.4.2.3.

3.6 Freedom from foreign matter. When tested as specified in 4.4.3, the fluid path of the syringe shall be free from chips, lint, dirt, or other foreign matter.

3.7 Safety (toxicity). When tested as specified in 4.4.4, the syringes, including tip protectors, shall comply with safety test requirements of the USP.

3.8 Lead content. When tested as specified in 4.4.5, all plastic and rubber components coming in contact with the medication being administered shall contain not more than 5 parts per million (ppm) of lead.

3.9 Freedom from pyrogens. When tested as specified in 4.4.6, syringes shall comply with the pyrogen test requirements of the USP.

3.10 Sterility. The syringe shall be sterile when tested with a biological indicator as specified in 4.4.7.

3.10.1 Certificate of sterility. A certificate of sterility stating that samples from each sterilizer lot have been tested and found to be sterile shall be submitted to the procuring activity. The certificate of sterility shall include:

a. Item identification.
b. National Stock Number (NSN).
c. Contract or purchase order number.
d. Sterility lot or control number.
e. Date of sterilization.
f. Quantity of product in sterilizer lot.
g. Type of sterilization (gas, radiation, heat, steam, etc.).
h. Form of biological indicator (i.e., inoculated paper spore strip or carrier, inoculated product, or inoculated simulated product).
i. Test organism(s) and population density used.
j. Test medium, incubation temperature, incubation time, and dates of tests.
k. Number of test units.
l. Statement that samples comply with sterility tests.
m. Signature of head of laboratory or a designated qualified alternate.

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When inspection is made at destination, the certificate of sterility shall be submitted by the contractor to the Defense Personnel Support Center, Directorate of Medical Materiel, ATTN: DPSC-ATQ, 2800 South 20th Street, Philadelphia, PA 19101, and a copy forwarded with each shipment to the consignee. When inspection is made at source, certificate of sterility shall be furnished to the cognizant Government Quality Assurance Representative for submission to the procuring activity.

3.10.2 Sterility lot. A sterility lot is that single quantity of product subjected to the same manufacturing operation and sterilized simutaneously in the same sterilizer chamber. Each unit package shall be clearly marked with a sterility lot (control) number and date of sterilization as specified in 5.3.1. Each sterility lot shall be traceable to the pertinent manufacturing lot(s).

3.11 Certificates. For destination inspection only, certificates of freedom from pyrogens and safety, as specified below, shall be submitted by the contractor to Defense Personnel Support Center, Directorate of Medical Materiel, ATTN: DPSC-ATQ, 2800 South 20th Street, Philadelphia, Pennsylvania, 19101, and one copy of each shall be forwarded with each shipment to the consignee.

3.11.1 Certificates of safety (toxicity). A certificate of safety, listing all lot or control numbers, shall state that samples from each lot or control have been tested for safety as specified in 3.7 and 4.4.4, and were found to comply with such requirements. The certificate shall reference the method used in determining safety.

3.11.2 Certificate of freedom from pyrogens. A certificate of freedom from pyrogens, listing all lot or control numbers, shall state that samples from each lot or control have been tested for pyrogens as specified in 3.8 and 4.4.5, and were found to comply with such requirements. The certificate shall reference the method used in determining freedom from pyrogens.

3.12 Instructions. Each unit package shall be provided with instructions describing the proper techniques for dispensing the sterile syringes from the containers and preparing them for use. In addition, wording shall be present instructing the user to destroy syringes after use to prevent their misuse.

3.13 Workmanship. Workmanship shall be first class throughout. Syringes (including components) shall be free from defects which detract from their appearance or impair their serviceability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturers of the rubber, polypropylene and silicone may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as indicated in table I. Unless otherwise specified, the unit of product for examination purposes shall be one syringe.

TABLE II. Sampling criteria for examination.

Type defect	Inspection level	AQL (percent defective)
Major A	S-4	1.0
Minor A	S-4	6.5
Minor B	S-1	4.0

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4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as indicated in table III. Unless otherwise specified, the unit of product for test purposes shall be one syringe. Where feasible, the same sample unit shall be used for the determination of one or more test characteristics.

TABLE III. Sampling criteria for tests.

Component	Characteristic	Require-ment	Test procedure	Inspection level	AQL (percent defective)
Barrel	Alcohol resistance	3.1.2.1	4.4.8	S-1	2.5
Plunger tip	Inertness	3.1.3.1	4.4.9	S-2	0.25
Syringe	Dimensions	3.2.1	-----	S-2	4.0
Syringe	Accuracy	3.3	4.4.1	S-1	2.5
Syringe	Pressure test	3.4	4.4.2.1	S-2	2.5
Syringe	Suction test	3.5	4.4.2.2 & 4.4.2.3	S-3 1/	1.5
Syringe	Foreign matter	3.6	4.4.3	S-2	4.0
Syringe	Safety	3.7	4.4.4	S-2	1.0
Syringe	Lead	3.8	4.4.5	S-2	1.0
Syringe	Freedom from pyrogens	3.9	4.4.6	S-2	1.0
Syringe	Sterility	3.10	4.4.7	S-2	1.0

1/ A minimum of 20 syringes from each lot shall be tested regardless of lot size.

4.3 Examination. The syringes shall be examined to determine compliance with all requirements contained in this specification. Examination shall be conducted as specified in tables II and IV. Examination, however, shall not be restricted to defects listed in table IV. Any syringes in the sample containing one or more defects shall be rejected and, if the number of defective syringes in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

TABLE IV. Classification of defects.

Categories	Defects
Critical	None defined.
Major A	
101	Tip protector missing or disloged.
102	Barrel not free from foreign matter.
103	Closed type protector not free of holes.

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TABLE IV. Classification of defects. - Continued

Categories	Defects
104	Tip protector frozen to syringe.
105	Plunger rod not straight.
106	Plunger rod or plunger tip missing.
107	Plunger, or plunger tip frozen in barrel.
108	Inside of barrel not coated with required lubricant.
109	Inside of barrel contains excessive lubricant.
110	Barrel tip orifice missing or blocked.
111	Graduation markings imcomplete, missing, illegible, or incorrect.
112	Barrel not clear, plunger tip not clearly visible making accurate alignment with graduation markings difficult or impossible.
113	Taper of barrel tip not smoothly finished nor free of pits, tool marks, burrs, easily detachable flash, or bubbles.
114	Inside surface of barrel not fine uniform surface, free from pits, grind marks, bubbles, mold marks, flash, open air lines, or other defects that would affect fluid tightness and smoothness of its fit with the plunger.
115	Plunger tip not smooth, free from flash, bubbles, breaks, cracks, or other defects affecting its fit with plunger.
116	Plunger does not operate smoothly in barrel.
117	Plunger tip detaches from plunger rod when pushed back and forth in syringe barrel.
118	Barrel or plunger (including tip) not free of cracks, breaks, crazing, or tackiness.
119	Immediate container fails to open readily.
120	Immediate container not free of holes, tears, cuts, or punctures.
121	Marking on immediate container missing, illegible incorrect, or incomplete.
Minor A	
201	Deep scratches (not interfering with reading of dosage, see 112) on outside surface of barrel (except on tapered portion of tip, see 113).
202	Contractor's name or registered trademark, missing, incomplete, or illegible.
203	Graduation scale not approximately in line with flattened side of flange.
204	Instructions missing, incomplete, or illegible.
Minor B	
251	(Unit of product - Unit package and contents) Average count per unit package not as listed in 6.2.

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4.3.1 Packaging inspection. The inspection of the unit package, packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking rquirements of MIL-STD-129.

4.4 Tests.

4.4.1 Accuracy test. Syringes shall be tested for accuracy at each main graduation mark (see table I) using water and a calibrated burette, or, as an alternate, gravimetrically, taking into account the temperature and density of water. The first ring from the distal end of the plunger tip shall be aligned with each graduation mark tested.

4.4.2 Fluid tightness test. The fluid tightness test shall consist of the pressure test and the suction test described in the following paragraphs:

4.4.2.1 Pressure test. The syringe shall be attached to a pressure testing machine, having a water pressure gage, through a connection with a suitable fluid path. Water, used as a testing fluid, shall be let in through the barrel tip until the pressure is raised to 50 pounds per square inch gage (psig). The plunger shall be drawn to a point half the graduated length where it shall be allowed to remain for 10 seconds. The plunger shall then be drawn back to midway between first position and the top graduation mark where it shall be allowed to remain for a similar length of time. The plunger shall then be drawn back to the top graduation mark where it shall be allowed to remain for a similar length of time. During each test, enough water shall not pass the connection between the testing machine and the barrel tip, or between the plunger tip and the barrel to produce a falling drop. After each successive test, any water that may have passed shall be wiped away and the next test applied.

4.4.2.2 Suction test. An amount of water equal to more than one-half the capacity shall be drawn into the syringe. The syringe shall then be held vertically, tip up, to allow air to rise above the water and be expelled. Tapping the barrel of the syringe will facilitate removal of trapped air. The plunger shall be pressed inward to the 50% mark. The tip shall be sealed. The plunger shall then be carefully withdrawn, at a rate of about one inch/10 seconds, to the 35 cc mark. After the plunger has reached the 35 cc mark, there shall be no bubbles coming by the distal ring for at least 10 seconds. Should there be more then three air bubbles passing the distal ring during the 10-second period, there shall be no bubbles passing the distal ring. Upon release of the plunger, it shall return to the 20 cc mark.

Note: Air bubbles which appear on the distal surface of the plunger tip during the withdrawal of the plunger shall be disregarded. Some air bubbles frequently appear under conditions of reduced pressure because of phenomena such as decreased solubility of air in water, interfacial tension, and aggregation of minute bubbles into larger ones. A continuing stream of bubbles arising from the plunger-barrel interface constitutes serious leakage.

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4.4.2.3 Alternate suction test. As an alternate, the suction test may be performed with a leak detector which is basically a thermistor conductivity meter using helium. Gas (helium) leakage past the plunger shall be indicated by deflections of the milliammeter needle as listed below for the syringe. Suction within the syringe shall be created by withdrawal of the plunger at a rate of approximately one-quarter inch per second. The thermistor conductivity meter which shall be used is the Gow-Mac Model 21-110, or similiar apparatus. To minimize hand operation a fixture shall be devised to hold the syringe, couple it to a suitable fitting, and work the plunger in a manner coincidental with the center line of the syringe barrel.

Sensitivity setting shall be low. A full scale deflection of the needle (50 divisions) for two or more seconds shall be considered as leakage and cause for rejection. A momentary full scale deflection shall not be considered an indication of leakage.

4.4.3 Freedom from foreign matter test. Syringe shall be tested as removed from individual container. The syringe shall be submitted to the tests described in the following paragraph:

4.4.3.1 Test method. Draw in an amount of distilled water equal to the capacity of the syringe. Eject the distilled water onto a medium texture, quantitative filter paper properly inserted in a 60°, 75 mm, smooth glass funnel. As a blank, take an equal amount of distilled water and pour it into a similar, separate funnel filter paper arrangement as used for the syringe. Examine the residue on the filter papers, after the water has passed through them, and determine if foreign material ascribable to the syringe is present.

4.4.4 Safety (toxicity). The finished syringes shall be tested for safety in accordance with the USP, for transfusion and infusion assemblies.

4.4.5 Test for lead. Ten (10) Gms of rubber or plastic shall be taken, ignited to char, and then further ignited in a muffle furnace at 700° C. to a white ash. The ash shall be boiled in 15 milliliters (ml) of 10 percent nitric acid, filtered into a volumetric flask and made up to 100 ml. An aliquot portion, equivalent to one gram of sample, shall be taken and placed in a separator. The dithizone test for lead, as given in the USP, shall be applied using a control of 5 ppm or 12 ppm of lead solution as given in the USP test. If the sample solution is darker than the limit, small aliquots shall be used until the color of the control is matched by the sample solution and the concentration of lead in ppm is determined.

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4.4.5.1 Alternate test for lead. Accurately weigh a 5-gram sample into a 1-oz. Vycor crucible. Reduce to ash in a muffle furnace at 650° C for eight hours. Let cool to room temperature and add 15 ml. of 1:1 hydrochloric acid. Bring to a boil on a hot plate, allow to boil for 2 to 3 minutes, remove, let cool, and filter through Whatman #2 filter paper into a 50-ml. volumetric flask. Rinse the residue on the filter with deionised water and dilute to 50 ml. Prepare standard lead solutions of 1, 2 and 5 ppm lead. Determine the absorbances of these solutions at the 283 millimicron wave length using a Hollow Cathode Pb lamp and a 10x scale expansion on a spectrophotometer, such as the Perkin-Elmer Model 303 Atomic Absorption Spectrophotometer. Determine the absorbance of the sample solutions using the same conditions as with the standards.

Calculation:

$$\frac{A \times F \times \text{dilution}}{W} = \text{ppm lead wherein,}$$

A = Absorbance
F = Factor = $\frac{\text{(Concentration standard in ppm Pb)*}}{\text{absorbance}}$
W = Sample weight in grams

* The three standards are averaged to give ppm/unit A.

4.4.6 Freedom from pyrogens. The pyrogen test shall be performed as described in the USP, for transfusion and infusion assemblies.

4.4.7 Sterility test. The syringe shall be tested for sterility in accordance with the method specified in the USP for parenteral devices using inoculated product, inoculated simulated product, or paper spore strip (carrier) 1/. Product containing a biological indicator shall be marked or labeled so that it clearly differentiates from product intended for distribution. The incubation medium and minimum time in each case shall be soybean casein digest for 7 days for inoculated product, inoculated simulated product, or paper spore strip (carrier). The incubation temperature shall be as indicated in paragraph 4.4.6.1. The minimum number of test units, regardless of lot size, shall be 10, except for inoculated carrier which shall be 20 for Type I Sterilization and 30 for Type II Sterilization including inoculated carrier and product test units. The inspection level shall be as indicated in table III, Sampling criteria for tests.

1/ The preinoculum shall be placed at those surfaces which in the intact article are least accessible to the sterilization medium. Sterilization shall be accomplished in the assembled state. Sterility testing shall be done in the dissassembled state.

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4.4.7.1 Inocualted product or simulated product. For each type of sterilization, the organism used shall be as listed below:

Type of Sterilization	Organism	Incubation Temperature
Steam	B. stearothermophilus	55° - 65° C
Dry Heat	B. stearothermophilus, or	55° - 65° C
	B. subtilis var. niger (or globigii)	30° - 35° C
Gas	B. subtilis var. niger (or globigii)	30° - 35° C
Radiation	B. pumilus	30° - 35° C

The number of viable cells shall be sufficient to provide assurance that the product has been sterilized.

4.4.7.2 Inoculated carrier. Inoculated carriers (paper spore strips) if used shall be in accordance with MIL-S-36586. Other paper carriers shall be inoculated to be of equivalent resistance. The method of incubation and number of test units shall be as indicated in 4.4.6 Sterility test and 4.4.6.1 Inoculated product or simulated product. The method of sterilization will control the one organism in the spore strip to be tested for as shown in 4.4.6.1. NOTE: When an inoculated carrier is chosen, product samples must also be tested as required by the USP.

4.4.7.3 Alternate method. Suppliers conducting sterility tests or designating lots in a manner other than that indicated herein shall submit proof that the alternate method provided equal assurance concerning the sterility of the product. However, all sterility tests shall be conducted using biological indicators. Requests for approval of any alternate method of testing for sterility shall be made to the Contracting Officer, and approval obtained before employment of the alternate method on a government contract.

4.4.7.4 Product testing. Testing for sterility of the product after receipt by the government shall be without a biological indicator and in accordance with the USP.

4.4.8 Permanency of graduation markings. The graduation markings shall withstand five vigorous rubbing cycles of a 70% isopropyl alcohol moistened cotton swab. A cycle shall be one up and one down stroke along the length of the barrel. The markings shall remain legible.

4.4.9 Plunger tip inertness. The rubber plunger tip shall be removed from the plunger rod, wiped with a clean, dry cloth to remove silicone lubricant, and immersed in a test tube under a volume of pH 10.5 buffer solution 1/ equal to the capacity of the syringe 2/. The exposure shall be for one hour at 68° to 77° F (20° to 25° C).

1/ The ph 10.5 buffer solution is prepared by dissolving 35.0 gm of dibasic potassium phosphate in 1000 ml of distilled water, and adding 2 ml of 10N potassium hydroxide. Adjust the pH with 18N phosphoric acid or 10N potassium hydroxide to 10.5 $\pm$ 0.1. The chemicals specified shall be ACS Reagent Grade quality.

2/ On large size plunger tips it is permissible to cut them into quarters and use only 1/4 volume of buffer solution.

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There shall be no color or any other visible change of the buffer solution. A control shall be used during the exposure consisting of the same amount of buffer solution in a test tube which does not contain a plunger tip.

It is permissible to cut the plunger tip into quarters and use only 1/4 the volume of sodium thiopental.

5. PACKAGING

5.1 Packaging. Packaging shall be level A or C as specified.

5.1.1 Unit of issue. One package containing 100 syringes as specified, constitutes one unit of issue.

5.1.1.1 Level A.

5.1.1.1.1 Immediate container. Each syringe shall be packaged in a readily opened container so constructed at to maintain the sterility of the syringe until manually opened.

5.1.1.1.2 Unit package. Each unit (100 immediate containers) shall be packaged in a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class domestic. Box shall be closed as specified in the appendix of PPP-B-636.

5.1.2 Level C. Syringes shall be packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A. Each unit packaged as specified in 5.1.1.1.2 shall be overpacked in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, class weather-resistant, V3c. Closure, waterproofing and strapping shall be as specified in the appendix of PPP-B-636.

5.2.2 Level B. No further overpacking is required. Unit package as specified in 5.1.1.1.2 shall be shipped as the exterior container.

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5.2.3 Level C. The packaged syringes shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier's rules and regulations.

5.3 Marking.

5.3.1 Immediate container. Each immediate container shall be commercially identified. Marking shall include the name of the contractor, lot (control) number, and the legends: "Sterile" and "Sterility Guaranteed Unless Package Is Damaged or Opened."

5.3.2 Unit package. Each unit package and exterior container shall be marked as specified in MIL-STD-129. When level B packing is specified, the unit package shall be marked as specified for the exterior container in MIL-STD-129. In addition, marking shall include the lot (control) number and the method of sterilization (G-Gas, R-Radiation, S-Steam, D-Dry Heat sterilization) shall be shown following the level of packaging and date of packaging. Example: "A-8/77-G." Final sterilization shall be on or after indicated packaging date.

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein, and the procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(c) NSN.

6.2 National stock number coverage. This specification is intended to cover the following item appearing in the Federal Supply Catalog:

National Stock Number	Item Identification
6515-00-727-0008	SYRINGE, IRRIGATING, SURGICAL, Plastic, Barrel and Plunger Type, Disposable, 50 cc, 2 fl. oz., 100s

6.3 Changes from previous issues. The asterisks to indicate changes have not been used in this specification due to the extensiveness of the changes.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activity:
DOD - MB

Preparing activity:
DLA-DM

Project No. 6515-2079

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-S-36876A SYRINGE, IRRIGATING, SURGICAL, CATHETER TIP, DISPOSABLE, 50 cc (2 Fl. Oz.)

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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